

Challenge (Habanero)

- Q1.** Solve for x : $2x + 5 = 11$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$
- Q2.** A gamer has 120 points. After a level, their points are given by $3x - 20$. If they now have 200 points, what is x ?
(A) $x = 20$
(B) $x = 30$
(C) $x = 40$
(D) $x = 50$
(E) $x = 60$
- Q3.** A coding error causes a program to use $2x + 10$ GB of data. If the program should use $5x - 20$ GB, what is x if $2x + 10 = 5x - 20$?
(A) $x = 5$
(B) $x = 10$
(C) $x = 15$
(D) $x = 20$
(E) $x = 25$
- Q4.** In a game, $3x - 2 = 5x + 1$. What is x ?
(A) $x = -1$
(B) $x = -0.5$
(C) $x = 0$
(D) $x = 0.5$
- (E) $x = 1$
- Q5.** Solve for y : $y - 3 = 2(y + 2)$
(A) $y = -1$
(B) $y = 0$
(C) $y = 1$
(D) $y = 2$
(E) $y = 3$
- Q6.** A scoreboard shows $x + 2 = 7$. What is x ?
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$
- Q7.** Solve for z : $z/2 + 2 = 6$
(A) $z = 4$
(B) $z = 6$
(C) $z = 8$
(D) $z = 10$
(E) $z = 12$
- Q8.** A gaming PC uses x GB of RAM. If $x - 4 = 7$, how much RAM does it use?
(A) $x = 3$
(B) $x = 7$
(C) $x = 11$
(D) $x = 15$
(E) $x = 20$

Open Ended Questions (FRQ)

Q9. Solve the equation $2x + 5 = 3x - 2$. Show all your work and check your solution.

Q10. The equation $x/4 + 2 = 9$ represents the number of lives a gamer has. Solve for x and explain the meaning of x in this context.

Chef's Tips

- Chef's Tips: Encourage students to read each question carefully and check their work.
- Remind students that showing their work is crucial for full credit on free-response questions.
- Emphasize the importance of using algebraic properties to simplify equations before solving them.